



College of Computing
Department of Computer Science and Artificial Intelligence



**وَالِي | WALI : AI-Based Proactive System for Preventing
Domestic Child Accidents**

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Abstract

Unintentional domestic accidents remain a significant threat to child safety, often occurring during brief moments of caregiver distraction. Existing monitoring solutions, such as traditional surveillance and baby monitoring systems, primarily provide passive observation without understanding the contextual relationship between children and surrounding hazards. This limitation highlights the need for a proactive and intelligent safety mechanism.

This project presents *WALI* , an AI-based proactive system designed to prevent domestic child accidents through real-time environmental monitoring. The system integrates Computer Vision techniques, including object detection, object tracking, and distance measurement, to identify both children and hazardous household objects (such as sharp tools, stair edges, swimming pool boundaries, and cleaning chemical storage areas) .A rule-based prediction approach is applied to analyze spatial and temporal relationships, enabling early detection of risky situations before direct interaction occurs.

The expected outcomes include high detection accuracy, low-latency alert generation, and reduced false alarm rates, ensuring practical usability in real-life environments. Ultimately, the system contributes to shifting home safety from reactive monitoring to proactive prevention by providing an intelligent, context-aware solution that enhances child protection and supports caregivers in making timely interventions.

Keywords: Computer Vision, Real-Time Object Detection, Proactive Hazard Detection, Domestic Child Safety, Injury Prevention.

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Glossary

Human-Centered AI (HCAI)	An approach to AI design that prioritizes human safety, ethics, and well-being, ensuring technology supports rather than replaces human supervision.
Contextual Risk Recognition	The process of analyzing the interaction and proximity between a child and hazardous objects to identify danger in real-time.
Proactive System	A system designed to detect and prevent potential accidents before they occur, rather than just recording them after the fact.
Artificial Intelligence (AI)	Technology that enables computers to perform tasks that normally require human intelligence, such as analyzing information and making decisions.
Computer Vision (CV)	A field of AI that allows computers to interpret and understand images or videos captured from the environment.
Object Detection Model	Computer vision technique is used to identify and locate objects within an image or video using bounding boxes.
Object Tracking	A technique used to follow the movement of detected objects across multiple video frames.
Distance Measurement	A method used to estimate how close the child is to a hazardous object based on their position in the image.
Rule-Based Prediction	A decision-making approach that applies simple logical conditions to determine whether a situation is potentially dangerous based on the relationship between detected objects.
YOLO	A real-time object detection algorithm used to detect and locate objects in images or video.
YOLOv5	A version of the YOLO model designed for fast and accurate object detection in computer vision applications.
YOLOv8	A newer version of the YOLO model that provides improved accuracy and performance for object detection tasks.
Jetson Nano	A small computing device developed by NVIDIA for running artificial intelligence and computer vision applications.
Eye Aspect Ratio (EAR)	A computer vision metric used to determine whether the eye is open or closed.
Raspberry Pi	A small, low-cost computer commonly used in embedded systems and (IoT) projects.
Internet of Things (IOT)	A network of connected devices that communicate and exchange data through the internet to enable monitoring, automation, and smart decision-making.
User Interface (UI)	the point of interaction between the user and the system, including elements like buttons, screens, and menus.

CHAPTER 1: INTRODUCTION & MOTIVATION

1.1 Context and background of the research problem

- *Project Overview*
- *The Global Crisis of Unintentional Injuries*
- *The Limitations of Traditional Supervision*
- *The Transition from Reactive to Proactive Technology*
- *The Smart Home Ecosystem*

1.2 Academic, Industrial, or Societal Relevance

1.3 Research Motivation and Significance

1.4 Expected Impact of the Study

1.5 Purpose of this Document

1.6 Project Goals and Objectives

1.7 Stakeholders

1.8 Scope, Constraints, and Assumptions

- *1.8.1 Scope*
- *1.8.2 Constraints*
- *1.8.3 Assumptions*

CHAPTER 1: INTRODUCTION & MOTIVATION

1.1 Context and background of the research problem

1.1.1 Project Overview

This project presents *WALI*, an AI-based proactive safety system developed to enhance child protection within domestic environments. By integrating Artificial Intelligence (AI) and Computer Vision (CV), the system is designed to continuously monitor the home environment in real-time, identify potentially hazardous situations, and deliver immediate alerts to caregivers before accidents occur.

Unlike conventional safety solutions that respond to incidents after they happen, *WALI* adopts a preventive approach by analysing the spatial relationship between children and surrounding household hazards. Through the application of object detection, object tracking, and distance measurement techniques, the system is capable of recognizing dangerous proximity patterns and generating proactive warnings, shifting the paradigm of domestic child safety from reactive monitoring to intelligent prevention.

1.1.2 The Global Crisis of Unintentional Injuries

Although the home is considered a haven, it can harbor numerous hazards for children. Due to their natural curiosity and lack of risk awareness, children are particularly susceptible to domestic accidents such as falls, burns, poisoning, and other unintentional injuries. These incidents raise serious global public health concerns. According to global health reports, unintentional injuries are one of the leading causes of death and disability among children worldwide. Recent estimates indicate that more than 305,000 children and adolescents died from unintentional injuries in 2021, emphasizing the severity of this issue [1].

Regional studies further validate the prevalence of these mishaps. For instance, a study conducted in Jeddah revealed that 54% of participating mothers reported that their children had experienced domestic injuries [2]. This trend is also reflected in Al-Baha, where research found that 58.2% of parents reported a history of domestic accidents among their children, with higher rates observed in larger households [3]. These statistics underscore the urgent need to address safety risks in domestic environments.

1.1.3 The Limitations of Traditional Supervision

Parental supervision is critical for preventing child injuries; however, maintaining constant attention in everyday life can be challenging. Parents and caregivers often experience what is commonly referred to as supervision fatigue, especially when managing multiple responsibilities within the household. Many domestic accidents occur within seconds during brief moments of distraction.

A child, for example, may attempt to climb a window, approach a staircase, or touch a hot stove while a caregiver answers the door or checks a mobile phone. These scenarios demonstrate the rapid and unpredictable nature of child behavior. As a result, human reaction time often fails to match the rapid onset of domestic unforeseen incidents, highlighting a fundamental flaw of relying solely on traditional supervision methods.

1.1.4 The Transition from Reactive to Proactive Technology

Traditional home safety tools are mostly reactive or passive. Reactive devices, such as child stair alerts, respond only after a child has already reached the edge. Passive safety measures, such as baby gates or cabinet locks, create physical barriers but lack the ability to actively monitor dynamic situations.

Recent advances in **AI** have created new opportunities for improving child safety through proactive monitoring systems. **CV** and real-time object detection technologies allow intelligent systems to analyze their surroundings and detect potentially dangerous interactions between children and hazardous objects. By continuously monitoring the environment, such systems can detect risk patterns and generate alerts before an accident occurs, shifting safety mechanisms from reactive responses to preventive interventions [4].

1.1.5 The Smart Home Ecosystem

The rapid growth of the Internet of Things (IoT) has transformed modern homes into interconnected smart environments in which devices communicate and exchange data to improve security, efficiency, and comfort. Within this evolving ecosystem, there is a growing expectation that smart technologies will play a proactive role in ensuring household safety.

In this context, the proposed system, **WALI**, serves as an intelligent safety layer within the home environment. By leveraging AI-based visual monitoring and real-time environmental analysis, **WALI** provides a sophisticated layer of protection designed to detect hazardous situations involving children and alert caregivers instantly. Through this integration, the system contributes to a larger vision of smart homes that actively protect vulnerable residents and improve overall child safety.

1.2 Academic, Industrial, and Societal Relevance

The significance of the **WALI** system is established through a comprehensive multi-dimensional impact, addressing critical gaps in current domestic safety frameworks:

- **Societal Relevance:** This research directly addresses a high-priority public health challenge related to the "Human Response Gap" in child supervision. Despite the presence of caregivers, domestic injury rates remain alarmingly high exceeding **54% in Jeddah [1]** and **58.2% in Al-Baha [2]**—primarily due to brief moments of distraction known as "supervision fatigue." The societal relevance of this project lies in its ability to provide a consistent, non-fatigable layer of protection. By utilizing automated real-time monitoring, the system offers a proactive social service that aims to reduce mortality and long-term disability rates among children. **Beyond its technical framework, WALI serves as a "silent guardian," bridging the gap between human limitations and the fundamental right of every child to a haven, ultimately providing families with an enduring sense of security and peace of mind.**
- **Academic Relevance:** In the academic sphere, this project contributes significantly to the fields of **AI** and **CV** by shifting the research focus from simple object recognition to **Contextual Activity Recognition and Spatial Analysis**. While standard models can identify a child or a hazard separately, the academic challenge addressed here is the real-time analysis of the interaction and proximity between them. This research provides a scholarly framework for solving technical issues such as minimizing system latency, optimizing real-time inference, and refining hazard prediction

algorithms to reduce false positives in dynamic domestic settings. It serves as a scientific reference for utilizing deep learning to create proactive, intelligent environments.

- **Industrial Relevance:** From an industrial perspective, the project aligns with the rapid expansion of the **Internet of Things (IoT)** and the global transition toward **Cognitive Smart Homes**. As the market for home automation evolves, there is a growing demand for devices that do more than just record video; they must "understand" and "act." *WALI* introduces a sophisticated safety layer that serves as a blueprint for the next generation of smart security products. By integrating AI with environmental analysis, the project provides a scalable industrial model that aligns with **Saudi Vision 2030's** goals for digital transformation [5], supporting the development of intelligent, safe, and technologically advanced living spaces that protect the nation's most vulnerable residents.

1.3 Research Motivation and Significance

The motivation behind the development of the *WALI* system is driven by the urgent need to evolve domestic safety from traditional methods to intelligent, autonomous frameworks. The following points highlight the core drivers and the overall significance of this research:

- **From Passive Monitoring to Proactive Intervention:** A major motivation is the technical limitation of current surveillance systems, which remain "passive" and dependent on continuous human oversight. The significance of *WALI* lies in its ability to transform these devices into "active agents" that can detect risks in real-time. By automating hazard recognition, we aim to bridge the gap between identifying a threat and preventing an accident before it occurs.
- **Mitigating Supervision Fatigue:** Human supervision is naturally subject to fatigue and inevitable moments of distraction. We are motivated to provide a "Technical Fail-safe" that acts as a non-fatigable guardian. The significance of this study is its potential to offer a consistent layer of protection that supports caregivers, ensuring that a few seconds of distraction do not lead to life-altering consequences.
- **Localization of Advanced AI Solutions:** While AI technology is advancing globally, there is a significant lack of specialized, locally developed systems that cater to the specific domestic environments within the region. We are motivated to pioneer a "Saudi-engineered" solution that aligns with local needs and cultural privacy, ensuring that the latest AI innovations are accessible and effectively integrated into households.
- **Advancing Contextual Risk Analysis:** From a research perspective, we are motivated by the challenge of moving beyond basic object detection. The significance of this project is its focus on "Spatial Risk Context"—analysing the relationship between a child and their surroundings. This contributes to the academic field by proving that AI can be trained to understand "danger" as a concept of proximity and interaction, not just as a static image.

The significance of this study extends beyond the code; it represents a strategic shift toward "Preventative Intelligence." By delivering a proactive system that combines global AI standards with local environmental understanding, this research offers a scalable model to reduce domestic accidents and alleviate pressure on healthcare systems. Ultimately, *WALI* aims to redefine the modern home as a safe, intelligent sanctuary through the power of purposeful innovation.

1.4 Expected Impact of the Study

The successful implementation of the *WALI* system is expected to generate a far-reaching impact across multiple domains, ranging from individual household safety to broader technological and economic contributions:

- **Social Impact: Redefining Child Safety and Well-being** The most immediate impact is the significant reduction in preventable domestic injuries. By providing a proactive "AI-guardian," *WALI* is expected to decrease the mortality and long-term disability rates associated with household accidents. Beyond the numbers, it offers a profound psychological impact by empowering caregivers and reducing "supervision anxiety." This fosters a more secure domestic environment, ensuring that the home remains a true sanctuary for child development and family peace of mind.
- **Technical Impact: Advancing Proactive Computer Vision** Technologically, this study is expected to set a new benchmark for **Contextual Risk Recognition** in private spaces. Unlike traditional motion-detection systems, *WALI* introduces a sophisticated framework for analysing spatial relationships and human-object interactions in real-time. This research provides a scalable technical blueprint for the next generation of "Aware" Smart Homes, proving that AI can evolve from a passive observer into an active, life-saving participant.
- **Economic Impact: Healthcare Optimization and Industrial Innovation** The adoption of such proactive systems is expected to yield substantial economic benefits by reducing both direct and indirect costs associated with domestic emergencies. By minimizing hospital admissions and long-term rehabilitation needs, the project contributes to the optimization of national healthcare expenditure. Furthermore, it acts as a catalyst for the **local AI economy**, showcasing how home-grown intellectual property can replace imported surveillance technologies and drive innovation in the high-value smart security market.
- **Academic Impact: Contributing to Human-Centric AI Research** On a scholarly level, this project contributes to the fields of **Artificial Intelligence** and **Digital Transformation** by providing a functional model of "**Human-Centric AI**" [6]. It serves as a scientific reference for future researchers exploring the ethics and efficiency of real-time hazard prediction. This study highlights the importance of tailoring global AI standards to meet specific domestic and environmental needs, paving the way for more specialized and purposeful AI applications that prioritize human safety and reliable interaction.

1.5 Purpose of this Document

The purpose of this document is to provide a comprehensive technical record of the analysis, design, and implementation of the AI-Based Proactive System for Preventing Domestic Child Accidents,

WALI. It explains the motivation behind the project, defines its objectives, and outlines the scope of the system, which focuses on detecting domestic hazards in home environments and providing real-time alerts to support proactive safety measures for children.

The document also discusses the methodology adopted for the system's development, including the design approach and technical components of the system. It serves as an official technical reference that justifies the selection of the AI algorithms, detection models, and evaluation metrics used in the project, particularly in relation to predictive analytics for real-time hazard prediction.

Finally, this document highlights the scientific and practical contributions of the system, demonstrating how it addresses the limitations of conventional home safety solutions by providing intelligent, proactive hazard detection with automatic intervention and alerting when necessary.

1.6 Project Goals and Objectives

WALI is developed with a set of well-defined goals that collectively aim to advance the state of domestic child safety through intelligent and proactive monitoring. The objectives of this project can be outlined as follows:

- **Proactive Hazard Detection:** The system is designed to simultaneously detect and localize both children and hazardous household objects — including sharp tools, stair edges, swimming pool boundaries, and cleaning chemical storage areas — within the monitored environment in real-time, enabling risk identification before any direct interaction occurs.
- **Spatial Risk Analysis:** Beyond simple detection, *WALI* analyzes the spatial relationship and movement trajectory between the child and surrounding hazards, assessing proximity trends and behavioral patterns to determine the likelihood of a dangerous encounter.
- **Early Warning Generation:** The system incorporates a rule-based prediction engine that evaluates detected risk conditions and generates proactive alerts with minimal latency, ensuring caregivers receive timely notifications that allow for immediate intervention.
- **High Detection Accuracy:** The project targets a high level of detection performance, measured through standard evaluation metrics including Accuracy, Precision, Recall, and mean Average Precision (mAP), to ensure that the system operates reliably across diverse real-world domestic conditions.
- **False Alarm Reduction:** A key objective is to maintain a low false alert rate, minimizing unnecessary interruptions while preserving the system's ability to detect genuine risks, thereby reducing caregiver alert fatigue and improving overall system trustworthiness.
- **Accessible Monitoring Interface:** *WALI* provides caregivers with an intuitive web-based dashboard that offers live camera feeds, real-time detection overlays, and prioritized alert notifications, ensuring that critical safety information is always accessible regardless of the device being used.

1.7 Stakeholders

WALI is developed with a clear understanding of the individuals and entities that are directly or indirectly affected by the system. The following stakeholders have been identified:

- **Caregivers and Parents:** The primary users of the system, who rely on *WALI* 's real-time alerts and live monitoring dashboard to supervise their children within the home environment. The system is designed to support their decision-making and reduce the cognitive burden of continuous manual supervision.
- **Children:** The core beneficiaries of the system, as *WALI* is fundamentally designed to ensure their safety and proactively reduce the risk of domestic accidents. While children do not interact with the system directly, their protection is the central objective driving every design and technical decision within this project.

- **Healthcare System:** An indirect stakeholder that stands to benefit from the anticipated reduction in emergency cases and hospital admissions resulting from preventable domestic injuries. By enabling early intervention, *WALI* contributes to alleviating pressure on healthcare resources and reducing the social and economic costs associated with child accidents.

1.8 Scope, Constraints, and Assumptions

1.8.1 Scope

The scope of this project includes the design and implementation of an AI-based proactive child safety monitoring system for indoor home environments. The system focuses on real-time detection of children and multiple household hazards using surveillance cameras and computer vision techniques. It also includes motion tracking, distance analysis, rule-based risk assessment, and a simple interface for caregivers to monitor alerts and live video feeds.

The project is specifically limited to selected domestic hazards that are commonly associated with child accidents, such as staircases, hot surfaces, sharp objects, cleaning products, and pool-related risks. The system is intended for indoor environments and does not include physical modifications to the home or integration with additional smart devices beyond the surveillance camera.

1.8.2 Constraints

Several constraints may affect the performance of the system, including variations in lighting conditions, camera positioning, field of view limitations, and partial occlusions within the scene. In addition, the real-time nature of video processing may impose computational constraints on system responsiveness and alert latency. The project is also constrained by the availability of suitable datasets representing real household hazards and realistic child–hazard interaction scenarios.

1.8.3 Assumptions

The system assumes that the surveillance camera is properly installed in an appropriate indoor location that provides sufficient visibility of both the child and the surrounding environment. It is further assumed that the selected hazards remain visible within the camera frame and that the monitoring environment supports continuous video streaming and stable system operation. It is also assumed that caregivers have access to the interface and can respond to generated alerts in a timely manner.

CHAPTER 2: BACKGROUND AND RELATED WORK

2.1 Relevant Concepts and Technologies

- *2.1.1 Artificial Intelligence (AI)*
- *2.1.2 Computer Vision (CV)*
- *2.1.3 Object Detection*
- *2.1.4 Object Tracking*
- *2.1.5 Distance Measurement*
- *2.1.6 Rule-Based Prediction*

2.2 Existing Systems or Solutions

- *2.2.1 Traditional Surveillance Systems*
- *2.2.2 Baby Monitoring Systems*
- *2.2.3 Research-Based Intelligent Monitoring Systems*

2.3 Comparative Analysis

2.4 Identified Gaps or Limitations

CHAPTER 2: BACKGROUND AND RELATED WORK

2.1 Relevant Concepts and Technologies

The proposed system relies on several key technologies that support intelligent monitoring and early detection of potential risks. This section briefly explains the role of each technology in the system, while detailed definitions of technical terms are provided in the **glossary section**.

2.1.1 Artificial Intelligence (AI)

AI serves as the core intelligence layer of *WALI*, enabling the system to go beyond simple video recording and perform meaningful analysis of the monitored environment. Through AI, the system can evaluate complex visual scenes, identify patterns of risk, and make real-time decisions regarding child safety. Rather than relying on continuous human supervision, AI allows the system to autonomously assess whether a child is exposed to a potential hazard and determine the appropriate response, thereby acting as a non-fatigable guardian within the domestic environment.

2.1.2 Computer Vision (CV)

CV is the primary perception mechanism through which *WALI* interprets and understands its surrounding environment. By processing live video frames captured from surveillance cameras, CV enables the system to recognize the presence of a child, identify hazardous household objects, and analyze spatial relationships within the scene. Unlike traditional motion-detection systems that only respond to movement, CV allows *WALI* to understand the context of what is happening in the environment, distinguishing between normal child activity and potentially dangerous situations that require immediate intervention.

2.1.3 Object Detection

Object detection techniques are used to identify the child and hazardous objects within the image and determine their locations using bounding boxes. This helps the system understand where the child and the potential danger are positioned in the scene.

2.1.4 Object Tracking

Object tracking is used to monitor the movement of the child across consecutive video frames. It allows the system to observe how the child moves over time and determine whether the child is approaching or moving away from a hazardous object.

2.1.5 Distance Measurement

Distance measurement is used to estimate the proximity between the child and the detected hazardous object. By calculating the distance between them, the system can determine how close the child is to a potentially dangerous area.

2.1.6 Rule-Based Prediction

Rule-based prediction applies predefined logical rules to identify potential risks before an accident occurs. For example, if the distance between the child and a hazardous object decreases continuously across several frames, the system may generate an early warning alert. This supports proactive safety by predicting possible danger situations.

2.2 Existing Systems

2.2.1 Traditional Surveillance Systems

Traditional surveillance systems are widely used in residential environments to provide general monitoring and security. These systems rely on CCTV cameras and motion detection to observe indoor activities and allow remote access through mobile or web applications. Some systems include features such as motion alerts, night vision, and cloud storage for recorded footage. However, these systems mainly provide passive monitoring, as they record or display video without understanding the context of the scene. They are unable to analyze environmental risks or detect hazardous situations, limiting their effectiveness in ensuring child safety [7].

2.2.2 Baby Monitoring Systems

Baby monitoring systems represent a class of smart home technologies that provide continuous supervision of infants and young children within domestic environments. These systems integrate real-time video streaming, audio communication, and advanced monitoring features to enhance parental awareness and support child safety.

- **Nanit Baby Monitor**

The Nanit Baby Monitor offers high-definition video streaming and two-way audio communication, complemented by features such as sleep pattern analysis, breathing monitoring, and automated notifications for unusual movements. By providing detailed insights into the child's activity and rest patterns, Nanit helps parents maintain informed oversight even when physically apart [8].

- **Owlet Smart Sock**

The Owlet Smart Sock is designed to monitor vital physiological indicators, including heart rate and oxygen saturation, while combining motion detection to deliver comprehensive safety updates. Alerts are transmitted to parents through a mobile application, enabling timely responses to potential concerns [9].

Despite these technological advancements, the primary emphasis of both systems remains on child-centered monitoring. While they provide valuable behavioral and physiological data, they offer limited assessment of environmental hazards that could pose risks within the home.

2.2.3 Research-Based Intelligent Monitoring Systems

Recent research studies have explored the use of AI and CV techniques to enhance children's safety through intelligent monitoring systems.

Zainab et al. [4] proposed a smart home monitoring system that utilizes object detection techniques to analyze video streams captured from indoor cameras. Several versions of the YOLO object detection model were evaluated to determine the most suitable model for real-time monitoring tasks. Based on

the experimental results, YOLOv5 was selected as the final model due to its high detection accuracy and real-time performance. The system detects children along with potential hazards such as fire or swimming pools and sends alerts when the child approaches dangerous areas. The system performance was evaluated using mean Average Precision (mAP50 and mAP50-95), Accuracy, and F1-score.

Singh et al. [10] introduced a monitoring system designed to improve infant safety during sleep using CV and AI techniques. The system analyzes video captured through a night-vision camera connected to a Jetson Nano processing device. It applies pose estimation techniques to analyze the infant's body posture and tracks eye movement using the Eye Aspect Ratio (EAR). When abnormal conditions are detected, alerts are sent to caregivers.

Li and Zhang [11] proposed a CV-based system to monitor children near swimming pools in order to reduce the risk of drowning. The system uses the YOLOv5 model to detect children and identify the pool area within the scene. When a child approaches the pool, the system sends alerts through a mobile application. The system integrates a surveillance camera connected to a small video processing device, an online platform for data management, and a mobile application for delivering notifications.

Zhang and Li [12] developed an intelligent mobile application that monitors children's safety inside homes using deep learning and object detection techniques. The system uses the YOLOv8 model to detect children and potentially hazardous objects such as doors and windows. By calculating the distance between the children and these objects, the system determines whether the child is approaching a risky area and sends alerts through a mobile application. The system was implemented using a Raspberry Pi connected to a camera along with an online platform for data management.

2.3 Comparative Analysis

Existing systems provide partial solutions for child safety, but they often focus on monitoring rather than analysing environmental risks. As illustrated in **Table 2.1**, a comparative analysis of existing approaches and the proposed *WALI* system reveals significant differences in terms of hazard detection, contextual awareness, and proactive alerting capabilities.

Feature	Traditional Surveillance Systems	Baby Monitoring Systems	Research-Based Systems	Proposed Systems (<i>WALI</i>)
Detects child presence	Yes	Yes	Yes	Yes
Detects environmental hazards	No	No	Limited	Yes
Understands context of danger	No	No	Limited	Yes
Provides early warning	No	No	Sometimes	Yes
Uses AI techniques	No	Limited	Yes	Yes
Combines child + hazard detection	No	No	Limited	Yes

Table 2.1: Comparative Analysis of Existing Child Safety Systems

2.4 Identified Gaps or Limitations

Existing systems provide useful features for monitoring children, but they still have several important limitations. Traditional surveillance systems mainly provide video recording and live monitoring, but they do not analyze the environment or detect potential hazards. These systems depend heavily on continuous human attention, which makes it difficult for caregivers to monitor the child at all times.

Baby monitoring systems often focus on physiological indicators such as heart rate, breathing patterns, or sleep tracking. Although these features are useful for health monitoring, they do not consider environmental dangers such as fire, sharp objects, stairs, or other harmful elements that may exist in the child's surroundings.

Some research-based systems apply AI techniques to detect specific risks, but they are often designed for limited scenarios or controlled environments. This makes them less flexible when applied to real-life situations where different types of hazards may exist simultaneously.

In addition, many existing solutions operate in a reactive manner, meaning that alerts are generated only after a dangerous event has already occurred. These systems typically detect the presence of the child or detect a single object, but they do not analyze the relationship between the child and the surrounding environment to predict potential danger in advance. Another limitation is that most existing approaches do not combine child detection with environmental hazard detection in a unified system. As a result, they may fail to provide a comprehensive understanding of the situation or support early risk prevention.

Therefore, there is a need for a proactive intelligent system capable of detecting both the child and environmental hazards simultaneously, analyzing their spatial relationship, and providing early warnings before accidents occur. The proposed system, *WALI*, aims to address these limitations by integrating real-time object detection, spatial risk analysis, and rule-based prediction within a unified proactive monitoring framework, thereby shifting domestic child safety from reactive responses to intelligent prevention.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Data Collection and Analysis

- *3.1.1 Data Source Selection*
- *3.1.2 Data Preprocessing*
- *3.1.3 Environmental Analysis*
- *3.1.4 Proposed Solutions*

3.2 User Interface (UI) Prototype

- *3.2.1 Prototype: Look and Feel.*
- *3.2.2 Live Monitoring and Alert Interface*
- *3.2.3 Web Push Notification*

3.3 System Requirements

- *3.3.1 Functional Requirements*
- *3.3.2 Non-functional Requirements*
- *3.3.3 Requirements Summary*

3.4 Data Analysis Overview

- *3.4.1 Data Processing Stages*
- *3.4.2 System Outputs*

3.5 Data Flow Diagram (DFD)

CHAPTER 3: SYSTEM ANALYSIS

3.1 Data Collection and Analysis

3.1.1 Data Source Selection

The proposed *WALI* system relies on visual data collected from both public datasets and custom domestic environment scenarios to support real-time child safety monitoring. The selected data focuses on detecting children and identifying hazardous household objects that may expose children to potential risks within indoor environments.

Publicly available datasets related to object detection and indoor monitoring are utilized to support the detection of common household hazards such as sharp objects, staircases, cleaning products, and swimming pool boundaries. In addition, custom image and video samples are collected to simulate realistic domestic scenarios involving child movement and environmental interactions. This combination helps improve dataset diversity and increases the system's ability to generalize across different home environments.

The collected data includes variations in camera angles, lighting conditions, room layouts, object positions, and levels of environmental complexity. Such diversity is important to ensure that the proposed system can operate reliably under realistic household conditions and support robust real-time hazard detection.

Particular attention is given to scenes containing multiple hazardous objects within a single camera frame, since the proposed system is designed to simultaneously analyze children and multiple surrounding hazards in real-time. This supports the development of a comprehensive intelligent monitoring framework capable of proactive safety analysis within domestic environments.

3.1.2 Data Preprocessing

Data preprocessing is considered a critical stage in preparing the collected visual data for real-time AI-based hazard detection and monitoring. The preprocessing pipeline is designed to improve data consistency, enhance detection performance, and support efficient model inference within dynamic domestic environments.

The preprocessing process begins with image and video annotation, where children and hazardous household objects are labeled using bounding boxes to support supervised object detection tasks. Hazard categories include dangerous tools, stair edges, cleaning chemical storage areas, and swimming pool boundaries. Annotation consistency is maintained to ensure accurate learning behavior during model training.

After labeling, all images and video frames undergo resizing and normalization procedures to standardize the input dimensions and pixel distributions required by the object detection model. These operations help improve computational efficiency and support stable real-time inference performance.

Additional preprocessing techniques are applied to improve model robustness under different environmental conditions. These include data augmentation operations such as brightness adjustment,

rotation, scaling, and horizontal flipping, which help simulate realistic domestic variations involving lighting changes, camera perspectives, and object positioning.

The dataset is then divided into training, validation, and testing subsets to support model development and performance evaluation. This structured preprocessing pipeline enables the proposed *WALI* system to efficiently analyze real-time video streams and accurately detect children and hazardous situations within complex household environments.

3.1.3 Environmental Analysis

The proposed *WALI* system aims to operate within a domestic environment that may contain several potential hazards for children. Therefore, environmental analysis is considered an important step in understanding unsafe situations and supporting proactive child safety monitoring before accidents occur. In this project, several common household hazards were selected based on their potential danger to children. These hazards include dangerous tools such as knives, forks, scissors, and iron, in addition to stair edges, swimming pool boundaries, and cleaning chemical cabinets. These objects and areas represent common sources of injuries and domestic accidents within residential environments.

1. Dangerous Tools

Dangerous tools such as knives, forks, scissors, and iron may expose children to injuries, cuts, or burns. The proposed system analyzes the spatial proximity between the child and these hazardous objects to identify potentially unsafe situations and generate early warnings when necessary.

One of the environmental challenges in this area is the presence of multiple surrounding household objects with visually similar shapes, which may make hazardous object detection more difficult.

2. Stair Edges

Stair edges represent dangerous areas where children may be exposed to falling accidents. The system monitors the child's proximity to stair boundaries to recognize unsafe situations and support proactive safety monitoring.

Variations in floor textures, stair designs, and lighting conditions may make stair boundaries less visually distinct in certain situations.

3. Swimming Pool Boundaries

Swimming pools are considered high-risk areas due to the possibility of drowning accidents. The system monitors the safety boundaries surrounding the pool area and identifies situations where the child approaches dangerous zones.

Outdoor lighting conditions and water reflections may create additional visual complexity around pool areas.

4. Cleaning Chemical Cabinets

Cleaning chemical storage areas may expose children to poisoning hazards. Therefore, the system monitors the child's proximity to these areas to identify potentially unsafe interactions.

These storage areas are often located in places that are easily accessible to toddlers and young children, increasing the possibility of accidental exposure.

By analyzing the relationship between the child and surrounding hazards, the proposed system aims to support intelligent and proactive child safety monitoring. Some areas within the home may be identified

as child safety monitoring. This helps the system recognize unsafe situations and generate early warnings before accidents occur, contributing to a safer domestic environment for children.

3.1.4 Proposed Solutions

The proposed *WALI* system aims to enhance child safety in home environments using artificial intelligence and computer vision technologies. Unlike traditional monitoring systems that are limited to passive observation, the proposed solution focuses on identifying potentially hazardous situations and generating early warnings before accidents occur.

The system is expected to monitor children and hazardous objects within the surrounding environment using deep learning-based object detection techniques. After detecting the child and nearby hazards, the system analyzes the spatial relationship and proximity between them to recognize unsafe situations.

The proposed solution focuses on several household hazards, including dangerous tools, stair edges, swimming pool boundaries, and chemical storage areas. By monitoring the child's position relative to these hazards, the system may support proactive safety monitoring and alert generation.

In addition, intelligent monitoring is supported by analyzing the child's movement and proximity to hazardous areas through sequential video frames. This may help identify situations where the child is approaching dangerous zones and allow the system to generate early warnings when necessary.

The proposed solution also includes a simple and user-friendly monitoring interface that may allow caregivers to observe the surrounding environment and receive safety alerts in real time. Overall, the system is expected to contribute to reducing domestic accidents and improving child safety through intelligent and proactive monitoring.

3.2 User Interface (UI) Prototype

The user interface for the *WALI* system is designed as a web-based platform to ensure accessibility and a seamless experience across various devices. The design focuses on simplicity and clarity, ensuring that the interface is intuitive and capable of delivering critical information effectively. By utilizing web-based architecture, the system ensures that caregivers can access the monitoring dashboard easily from any browser, facilitating a fast response in emergency situations.

3.2.1 Prototype: Look and Feel

The visual identity of *WALI* is based on a **Soft Beige**, **Gold**, and **Navy Blue** color palette. Beige tones create a calm and visually comfortable environment for users, while gold accents highlight key elements and reflect the value of child safety. Navy blue is used to reinforce a sense of security, trust, and system reliability. Together, these colors help critical alerts stand out clearly within a clean and minimalist interface design.

The following figure showcases the main dashboard, which serves as the primary entry point for monitoring. As shown in **Figure 3.1**, it centralizes all essential tools and features as detailed below:

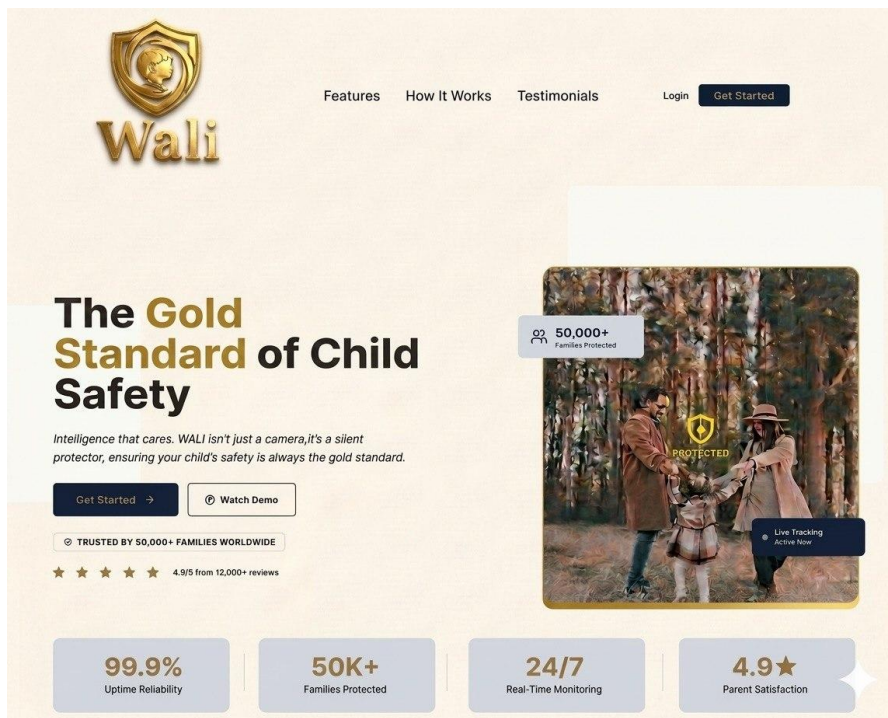


Figure 3.1: *WALI System Landing Page and Visual Identity*

The following interface highlights the core features of the **WALI** system, including intelligent safety monitoring, hazard alerts, and privacy-focused local AI processing. **Figure 3.2** demonstrates how these features are organized within a simple and user-friendly layout for caregivers.



Figure 3.2: *Core System Features*

The following interface demonstrates the operational workflow of the **WALI** system in three simplified steps. As illustrated in **Figure 3.3** caregivers begin by setting up the monitoring device, defining hazard zones within the environment, and then receiving real-time alerts and live monitoring updates through the system interface. The layout is designed to present the workflow in a clear and intuitive manner for ease of use.

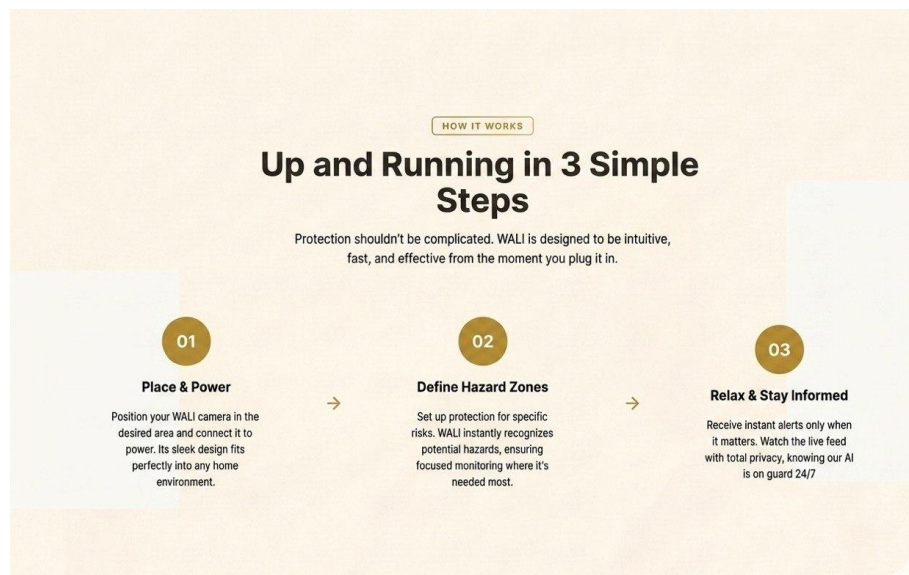


Figure 3.3: System Operational Workflow

The following interface presents user experiences and trust indicators within the **WALI** system. As demonstrated in **Figure 3.4** the interface highlights caregiver feedback regarding the system's monitoring and alert capabilities. It also presents key security and privacy features, including encryption, privacy protection, compliance standards, and continuous support, reinforcing the system's reliability and trustworthiness.



Figure 3.4: User Testimonials and Feedback Interface

The following interface presents the final section of the **WALI** website interface. As illustrated in **Figure 3.5** the interface provides users with access to essential system information, including navigation links, privacy policies, contact details, and support resources. The design maintains a clean and visually consistent layout that reinforces the system’s focus on safety, trust, and accessibility.

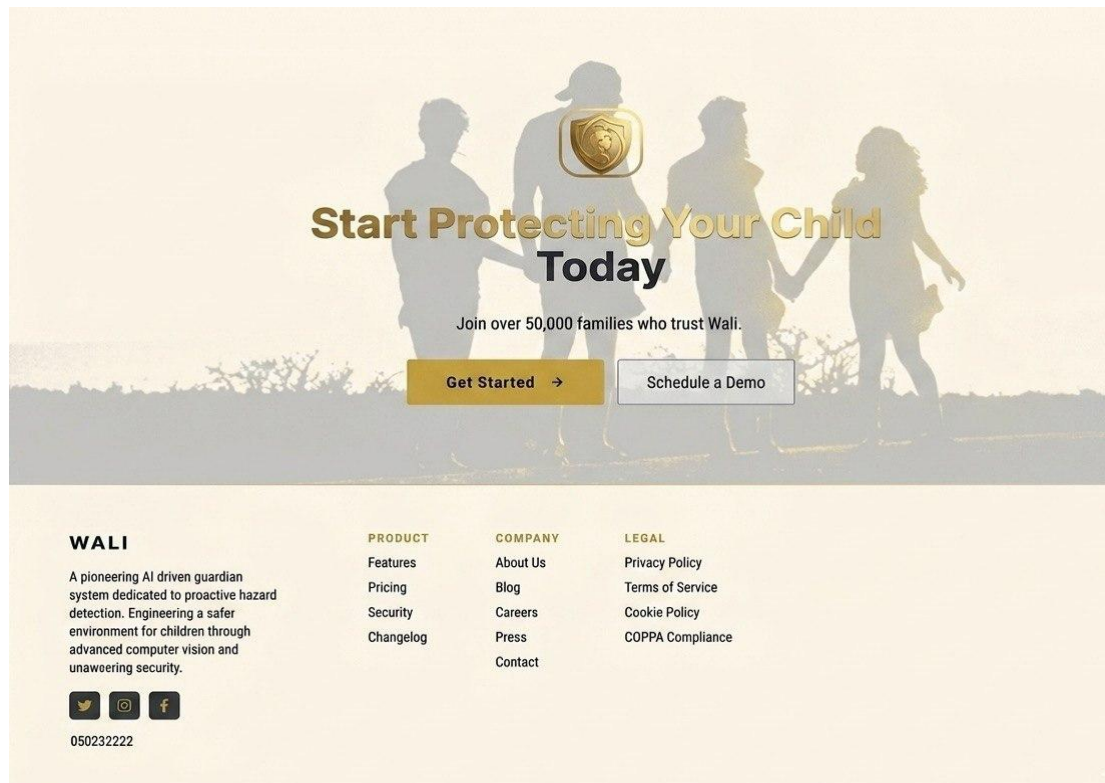


Figure 3.5: *System Onboarding and Contact Information*

3.2.2 Live Monitoring and Alert Interface

This interface integrates real-time video surveillance with an intelligent notification system to ensure continuous safety, as illustrated in **Figure 3.6**. The layout is divided into several functional zones:

- **Live AI Feed:** Displays the camera stream with active bounding boxes that identify hazards (e.g., stairs) and track the child’s proximity to risks.
- **Prioritized Alerts:** A dedicated sidebar categorizes detections into three severity levels (High, Medium, and Low Risk) enabling immediate response to critical situations.

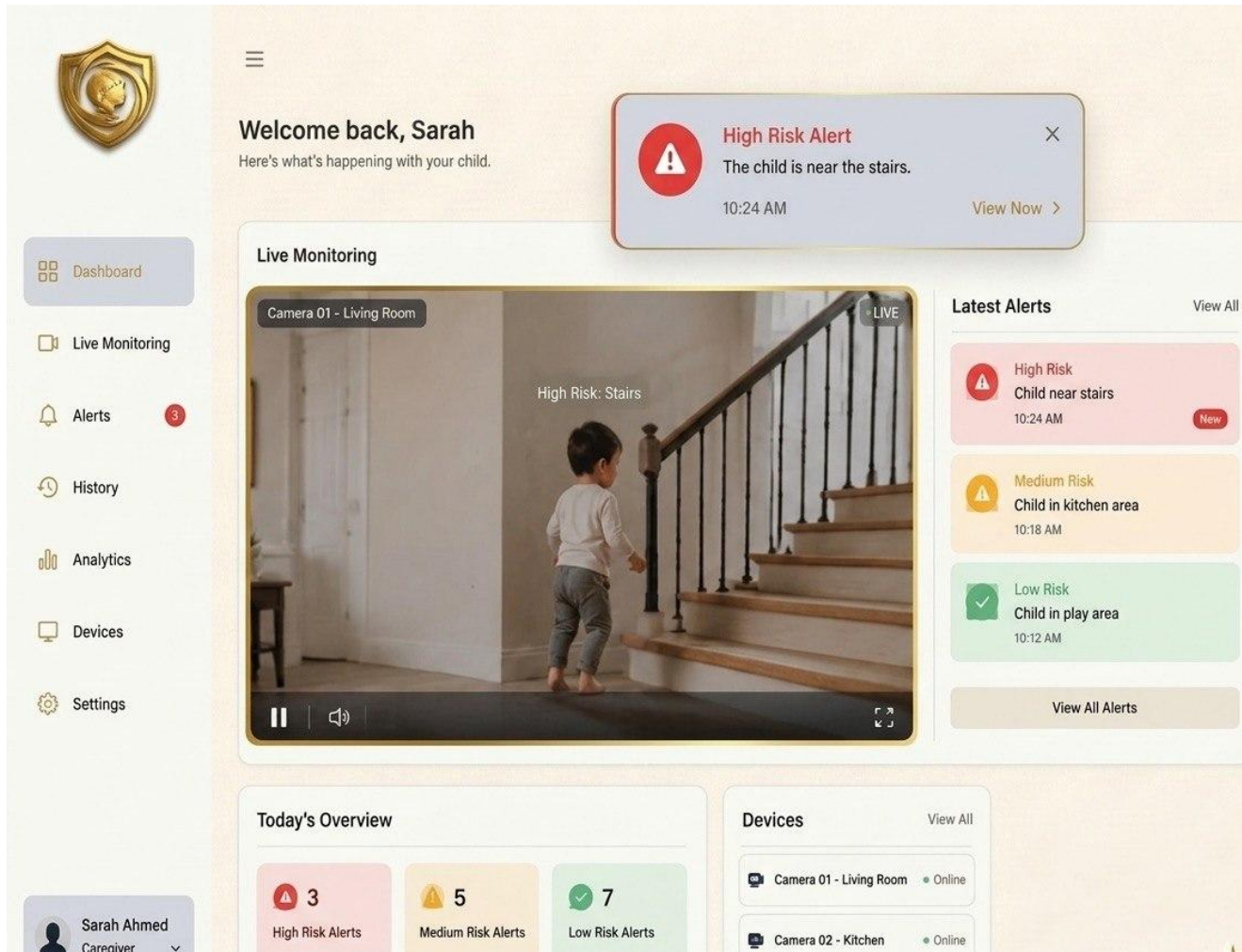


Figure 3.6: *Integrated Live Monitoring Dashboard with Alert*

3.2.3 Web Push Notification

To ensure maximum responsiveness, **WALI** includes a web push notification system designed for immediate reach. The notification interface presented in **Figure 3.7** demonstrates the system's real-time alert functionality:

- **Real-time Alerts:** Instant push notifications that appear on the user's lock screen, providing a concise description of the detected hazard.
- **Urgency Levels:** Notifications are categorized by risk severity, using clear language to prompt immediate parental action.
- **Cross-Platform Accessibility:** The notification system is engineered to be compatible with both **iOS and Android** ecosystems, ensuring all users receive critical updates regardless of their device.

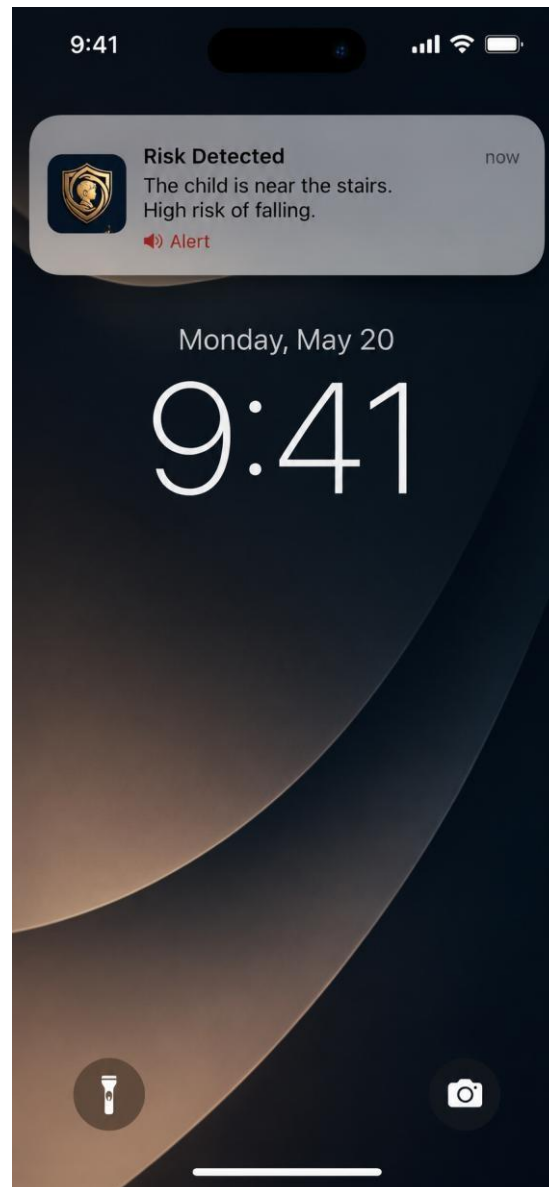


Figure 3.7: *Web Push Notification Interface*

3.3 System Requirements

The system requirements specify the essential services, constraints, and quality attributes that the proposed *WALI* system must meet in order to provide consistent, proactive child-safety monitoring in domestic settings. Requirements are divided into two types: functional requirements (**what the system should do**) and non-functional requirements (**how well the system should perform and under what constraints**).

3.3.1 Functional Requirements

The functional requirements listed below describe the system's core operational capabilities.

FR-01 Real-Time Video Monitoring

The system shall incessantly receive and process live video streams from connected surveillance cameras for indoor environment monitoring.

FR-02 Child Detection

The system shall autonomously detect the presence of a child within the monitored scene using Computer Vision techniques.

FR-03 Hazard Detection

The system shall autonomously detect the predetermined household hazard classes: dangerous tools (knife, scissors, iron), stair edges, swimming pool boundaries, and cleaning chemicals.

FR-04 Multi-Object Detection

The system shall concurrently detect and localize both the child and hazardous objects within the same video frame.

FR-05 Object Tracking

The system shall track the child's movement across sequential frames to determine motion direction, speed, and projected path.

FR-06 Proximity Measurement

The system shall calculate the closeness distance between the child and detected hazards in real time.

FR-07 Risk Assessment

The system shall appraise contextual risk based on proximity, movement behavior, hazard type, and predefined safety rules

FR-08 Predictive Alert Generation

The system shall produce an early warning alert when a high-probability hazardous encounter is predicted before direct contact occurs.

FR-09 Notification Delivery

The system shall inform to caregivers through the system dashboard and supported notification channels (web notification).

FR-10 Live Monitoring Dashboard

The system shall provide an interface displaying live camera feeds, detection overlays, risk indicators, and active alerts.

FR-11 Event Logging

The system shall archive significant events, including detected risks, generated alerts, timestamps, and camera source references.

FR-12 User Authentication

The system shall confine dashboard access to authorized users through secure identifiers (two-factor authentication).

FR-13 Camera Management

The system shall allow authorized users to register, remove, and configure surveillance cameras. A camera shall be connected to the local network and provide an accessible video stream prior to being added to the system.

FR-14 System Availability Handling

The system shall alert users when camera feeds are disconnected or unavailable.

FR-15: Create Account

The system shall enable caregivers to register a new account securely before accessing system services.

3.3.2 Non-functional Requirements

The non-functional requirements listed below specify system quality attributes and operational constraints.

A. Performance

NFR-01 Alert Response Time

The system shall trigger and display alerts within 2 seconds or less from the moment a high-risk condition is detected.

NFR-02 Real-Time Processing

The system shall process video streams at a consistent operational frame rate adequate for real time monitoring.

NFR-03 Concurrent Monitoring

The system shall support simultaneous monitoring of multiple camera feeds, subject to available hardware resources.

B. Accuracy

NFR-04 Detection Accuracy

The object detection model shall achieve a target mean Average Precision (mAP) of not less than 0.90 under standardized evaluation conditions.

NFR-05 False Alarm Control

The system shall mitigate false positives and false negatives to maintain practical reliability and reduce alert fatigue.

NFR-06 Resilience

The system shall maintain acceptable detection performance under normal variations in lighting, partial occlusion, and camera angle changes.

C. Security and Privacy

NFR-07 Local Data Processing

Where deployment resources permit, video analysis shall be processed locally to reduce privacy risks and external data exposure.

NFR-08 Secure Communication

All transmitted camera streams, alerts, and user sessions shall adhere to secure communication mechanisms. Like HTTPS

NFR-09 Access Control

The system shall mandate authorization for all personnel by accessing sensitive data and system settings.

NFR-10 Data Retention Control

Stored logs and event records shall be manageable according to configurable retention policies.

D. Reliability

NFR-11 Continuous Operation

The system shall be capable of stable long-duration operation for daily home monitoring scenarios.

NFR-12 Fault Tolerance

Temporary interruptions in camera feeds or network connectivity shall be detected and reported without causing full system failure.

E. Scalability

NFR-13 Extensibility

The system architecture shall support future integration of additional cameras, new hazard classes, and advanced notification channels.

NFR-14 Modular Upgradeability

AI detection models and rule sets shall be replaceable or updatable without redesigning the full system.

3.3.3 Requirements Summary

The requirements outlined above ensure that *WALI* functions as a proactive domestic child safety system that is accurate, responsive, secure, scalable, and user-friendly. These requirements also establish measurable criteria for the subsequent system design, implementation, and testing phases.

3.4 Data Analysis Overview

The *WALI* system transforms raw environmental observations into meaningful safety decisions. The process involves analyzing real-time data captured from the home environment to identify the child, detect hazardous household elements, and assess potentially dangerous situations.

3.4.1 Data Processing Stages

The data processing within the *WALI* system, as shown in **Figure 3.8** is executed through three sequential stages that transform raw environmental input into actionable safety insights:

- **Raw Visual Data:** This initial stage involves the continuous capture of the live video stream from the home environment. The system extracts specific image frames to be prepared for high-precision analysis.

- **Detection Data:** In this phase, the AI/CV detection module identifies the child’s presence and detects nearby household hazards, such as staircases, swimming pools, or other risky zones, within the monitored scene.
- **Analytical and Decision Data:** The final stage performs a contextual analysis of the detected objects. It evaluates the child’s proximity to hazards, movement direction, and spatial relations to determine the risk level and decide if the situation represents a potential danger.

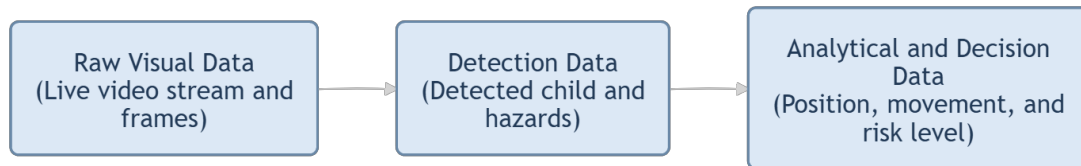


Figure 3.8: Main data categories handled by the proposed *WALI* system

3.4.2 System Outputs

The system generates several outputs including the live camera feed, detected hazard information, and real-time warning notifications delivered through the user interface to support proactive intervention.

3.5 Data Flow Diagram (DFD)

The **Context Level Diagram**, as illustrated in **Figure 3.9**, defines the scope of the *WALI* system. It shows the system as a single process interacting with two external entities: the **Camera**, which provides the input stream, and the **Caregiver**, who receives processed safety information and alerts. This high-level view ensures all external data interfaces are identified before diving into internal system details

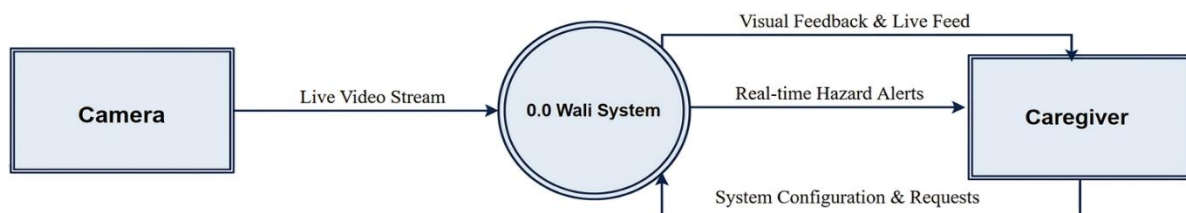


Figure 3.9: Context Level DFD for *WALI* System

Figure 3.10 presents the System **Level 1 DFD** of the *WALI* system. The diagram illustrates the internal data flow between video stream preprocessing, AI inference and feature extraction, and the event-driven alerting process. It also demonstrates the interaction between the caregiver, system alerts, and stored user preferences within the monitoring workflow.

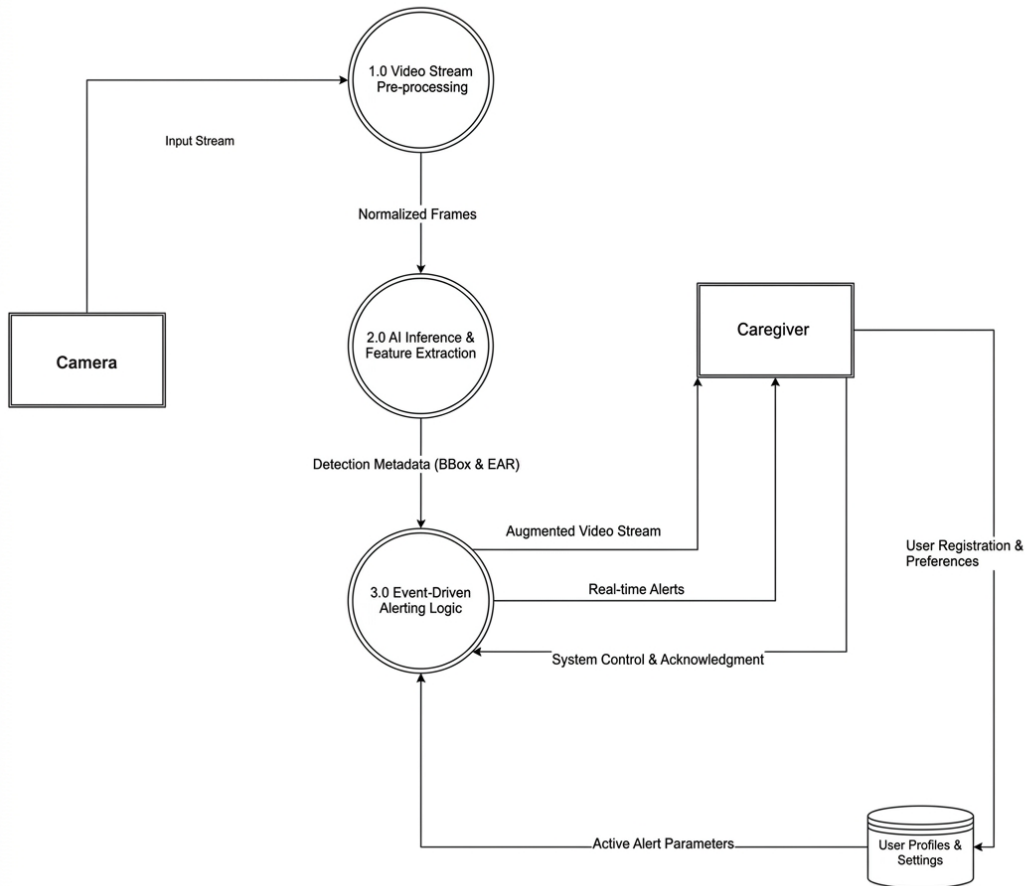


Figure 3.10: *System Level 1 DFD*

Process Descriptions:

- **Process 1.0 (Video Stream Pre-processing):** Standardizes the incoming raw video stream into optimized frames, ensuring consistent data quality for the analysis phase.
- **Process 2.0 (AI Inference & Feature Extraction):** Executes automated detection to identify potential hazards and monitor safety metrics from the processed visual data.
- **Process 3.0 (Event-Driven Alerting Logic):** Evaluates the detection results against active user parameters to trigger precise, event-driven notifications.

CHAPTER 4: SYSTEM MODELING TOOLS

4.1 UML Diagrams

- *4.1.1 Use case Diagram*
- *4.1.2 Use Case Table*

4.2 Sequence Diagram

4.3 Activity Diagram

CHAPTER 4: SYSTEM MODELING TOOLS

4.1 UML Diagrams

4.1.1 Use Case Diagram

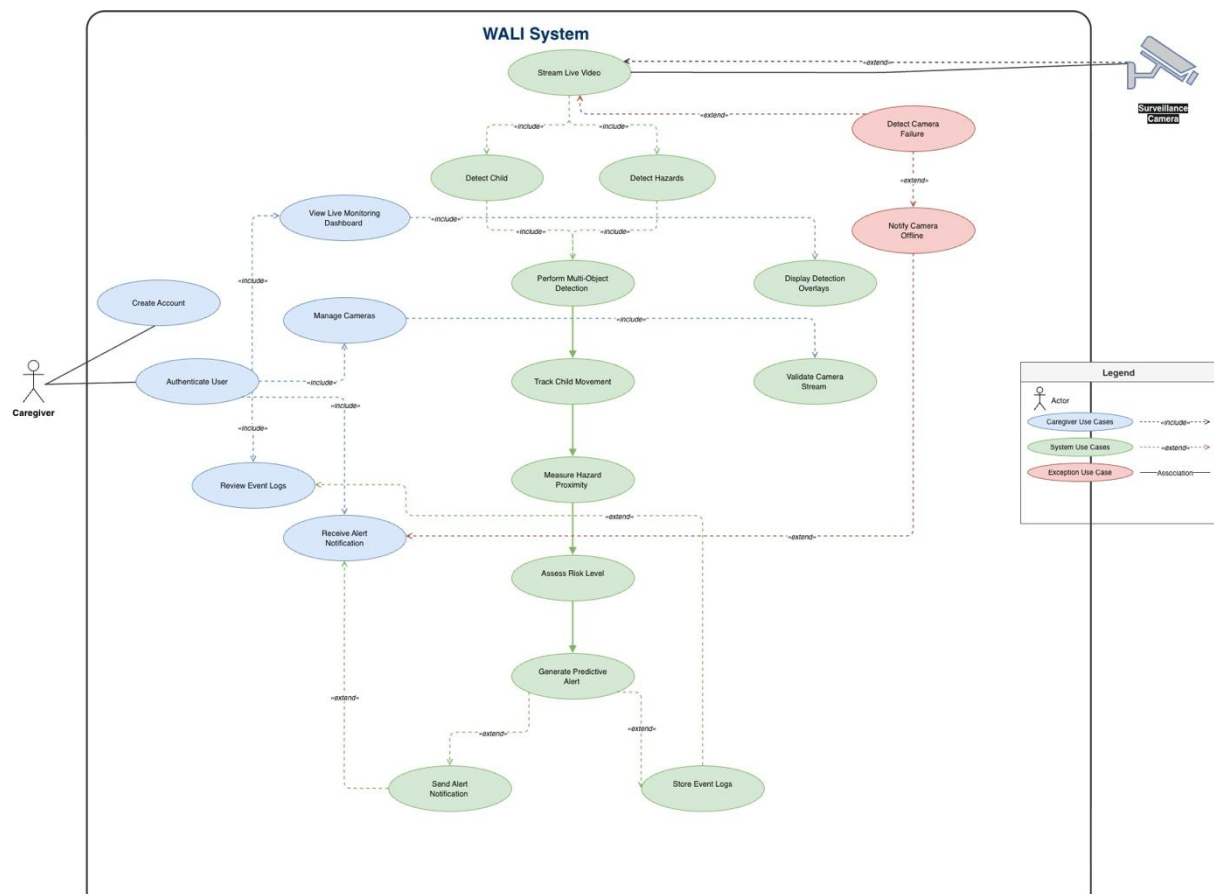


Figure 4.1: Visualising Interactions Between the Caregiver, The AI System, And the Camera

4.1.2 USE CASE Table

Phase 1: Access Control

This phase covers the initial interaction of the caregiver with the system, focusing on account creation and security, and as illustrated in **Table 4.1** and **Table 4.2**, the system handles both sign-up and authentication scenarios.

Sign-Up Scenario	
Actor	Caregiver.
Description	1. The user opens the sign-up screen. 2. The user enters (First Name, Last Name, Phone Number, Email, and Password). 3. The system validates the data and ensures the phone number/email are unique. 4. The account is created and stored in the database. 5. The system displays the home page/dashboard.
Data	First Name, Last Name, Phone Number, Email, Password.
Stimulus	When the user opens the sign-up screen for the first time.
Response	Create a new user account and grant access to the application.

Table 4.1: Use Case Scenario for Sign Up

Authenticate Scenario	
Actor	Caregiver.
Description	1. The user accesses the login screen. 2. The user enters (Phone Number, Email, and Password). 3. The system validates the credentials against the user database. 4. The user is redirected to the Monitoring Dashboard.
Data	Phone Number, Email, Password.
Stimulus	When the user opens the application login screen.
Response	Grant access to the system features and dashboard.

Table 4.2: Use Case Scenario for Authenticate User

Phase 2: Intelligent Monitoring

This phase details the real-time monitoring process, ensuring system stability and automated object recognition, and as illustrated in **Table 4.3** and **Table 4.4**, the system manages both connection status feedback and multi-object detection.

Use Case Scenario for Connection Status Feedback	
Actor	System.
Description	1. The system monitors the heartbeat of the camera connection. 2. If the signal drops, the system displays a "Camera Offline" warning. 3. An emergency notification is sent: "Monitoring Interrupted." 4. The caregiver is prompted to check the hardware or network.
Data	Connection Heartbeat, Latency.
Stimulus	Sudden drop in video stream data.
Response	Visual and audible warning to the user that monitoring is inactive.

Table 4.3: *Use Case Scenario for Connection Status Feedback*

Use Case Scenario for Multi-Object Detection	
Actor	System.
Description	1. The system streams live video from the camera. 2. Detect child and hazards in the frames using AI models. 3. Track movement and measure distance (Proximity). 4. Display detection overlays (bounding boxes) on the screen.
Data	Live Video Stream, Object Coordinates, Distance Metrics.
Stimulus	When the camera stream is activated and validated.
Response	Visual detection of hazards and real-time tracking on the dashboard.

Table 4.4: *Use Case Scenario for Multi-Object Detection*

Phase 3: Proactive Safety

This phase describes the transition from monitoring to active protection, focusing on predictive alerting and immediate caregiver notification, and as illustrated in **Table 4.5** and **Table 4.6**, the system handles both predictive alerting and caregiver feedback scenarios.

Use Case Scenario for Predictive Alerting	
Actor	System.
Description	1. Assess the risk level based on the calculated proximity. 2. Trigger an immediate predictive alert if a safety threshold is violated. 3. Send a real-time push notification to the caregiver's device.
Data	Risk Level, Notification Message
Stimulus	When the child approaches a pre-defined hazard zone.
Response	Immediate notification to the caregiver for intervention.

Table 4.5: *Use Case Scenario for Predictive Alerting.*

Use Case Scenario for Caregiver Feedback	
Actor	Caregiver / System.
Description	1. The system provides continuous visual feedback via Display Detection Overlays on the live stream. 2. When a hazard is detected, the system issues an immediate audible or visual alert. 3. The caregiver receives the Alert Notification on their interface. 4. The caregiver intervenes based on the real-time feedback provided.
Data	Visual Overlays, Alert Status, Notification Message.
Stimulus	When the AI model detects a change in the safety status.
Response	Immediate awareness for the caregiver to ensure the child's safety.

Table 4.6: *Use Case Scenario for Caregiver Feedback.*

Phase 4: Continuous Improvement

This phase focuses on the system's ability to learn from user feedback, ensuring long-term accuracy through data-driven model refinement, and as illustrated in **Table 4.7**, the system collects caregiver feedback following each detection event.

Use Case Scenario for Model Performance Feedback	
Actor	System.
Description	1. After an alert is triggered, a quick feedback prompt appears (e.g., "Was this alert accurate?"). 2. The caregiver selects (Yes / No) or provides a rating. 3. The system captures the detection metadata associated with the feedback. 4. The data is sent to the server for model retraining and accuracy analysis.
Data	Accuracy Rating (Thumbs up/down), Model Metadata, Timestamp.
Stimulus	Following a detection event or an alert notification.
Response	Submission of performance data to improve model precision.

Table 4.7: *Use Case Scenario for Model Performance Feedback*

4.2 Sequence Diagram

The Sequence Diagram illustrates the interaction flow between the main components of the proposed **WALI** system during real-time hazard monitoring and alert generation. It describes the chronological sequence of operations starting from video capture until the caregiver receives a safety notification.

The monitoring process begins when the surveillance camera continuously captures live video frames from the home environment and sends them to the AI-based detection module. The detection module processes the incoming frames using Computer Vision techniques to identify the child and surrounding household hazards.

After object detection is completed, the detected information is forwarded to the motion tracking and risk analysis module. This module tracks the child's movement across consecutive frames and calculates the spatial distance between the child and hazardous areas or objects. The rule-based prediction mechanism then evaluates the child's movement direction, proximity trend, and potential risk level.

If the analyzed situation satisfies the predefined danger conditions, the system generates an early warning alert. The generated alert is transmitted to the user interface, where caregivers receive real-time notification together with the monitored camera feed and detected hazard information.

The Sequence Diagram in **Figure 4.2** therefore demonstrates how the proposed system transforms live environmental observations into proactive safety alerts through coordinated interaction between detection, tracking, analysis, and notification components.

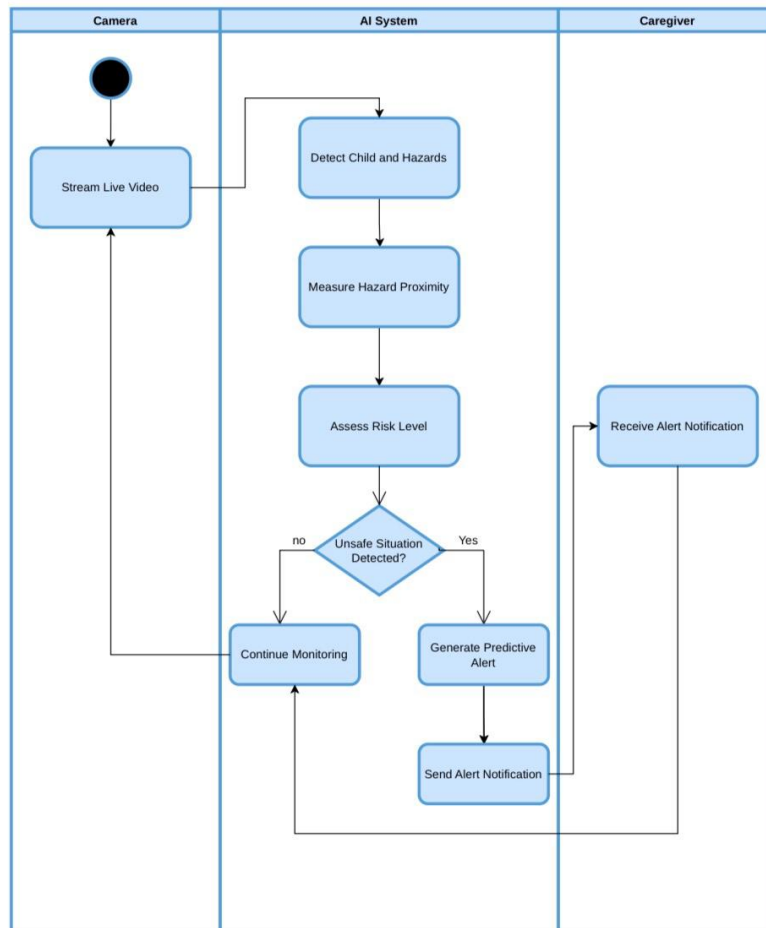


Figure 4.3: Activity Diagram of the *WALI* System

CHAPTER 5: DESIGN CONSIDERATIONS

5.1 Tools and Techniques

- *5.1.1 Development Tools*
- *5.1.2 AI Techniques*

5.2 Architectural Strategies

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CHAPTER 5: DESIGN CONSIDERATIONS

5.1 Tools and Techniques

The *WALI* system's development required the integration of multiple hardware and software technologies to support real-time monitoring, intelligent hazard recognition, and proactive child safety analysis. Because the proposed system operates in dynamic domestic environments, the tools and techniques used were carefully chosen to ensure high detection accuracy, low-latency processing, scalability, and consistent real-time performance.

The technical framework combines **Artificial Intelligence (AI)**, **Computer Vision (CV)** real-time object detection, object tracking, and spatial risk analysis to transform traditional surveillance into a proactive, intelligent safety solution.

5.1.1 Development Tools

The design of the *WALI* system incorporates a carefully selected suite of hardware and software technologies. These tools have been strategically chosen to ensure the system's ability to perform intelligent video processing, low-latency AI inference, and proactive hazard prediction once implemented.

Hardware Architecture Design

The proposed hardware infrastructure is designed to facilitate continuous monitoring and high-speed execution of detection models. The system will leverage surveillance cameras for real-time video acquisition within domestic environments.

To achieve Edge AI capabilities, the design specifies the use of compact computing units such as the **NVIDIA Jetson Orin Nano**. This choice is motivated by its integrated GPU architecture, which is essential for accelerating **YOLO inference at the edge**. For the model development and training phase, a high-performance workstation equipped with a dedicated GPU will be utilized to handle the intensive computational requirements of deep learning.

Software Ecosystem and Frameworks

The *WALI* system will be developed using modern frameworks that prioritize scalability and real-time performance:

- **Python:** Chosen as the core programming language for its robust ecosystem in AI and its efficiency in handling asynchronous data streams.
- **YOLO (You Only Look Once) Object Detection Framework:** The system will employ the latest stable iteration of the YOLO framework as its core object detection engine. YOLO was selected for its state-of-the-art performance in real-time environment sensing, providing an optimal balance between mean Average Precision (mAP) and inference speed. By utilizing this framework, the system will achieve simultaneous detection and localization of children and potential hazards with high computational efficiency, ensuring rapid response times in domestic safety scenarios.

- **OpenCV:** Planned for the image processing pipeline, specifically for frame transformation, spatial coordinate calculations, and real-time visualization of safety boundaries.
- **PyTorch:** *Will* serve as the primary deep learning library to manage neural network operations and optimize GPU memory utilization.
- **Web-Based Interface:** The system design includes a web platform to ensure cross-device accessibility. This interface will be responsible for presenting live telemetry and delivering immediate alerts to caregivers.

Tool / Technology	Purpose
Python	Main programming language
YOLO Framework	Real-time object detection and spatial localization
OpenCV	Video processing and spatial calculations
PyTorch	Deep learning framework and model management
Jetson Orin Nano	High-performance Edge AI processing
Web Platform	Real-time monitoring and alert interface

Table 5.1: *Summary of Development Tools*

5.1.2 AI Techniques

The proposed **WALI** system integrates several artificial intelligence and computer vision techniques to support intelligent and proactive child safety monitoring within domestic environments.

The system primarily relies on **Object Detection** techniques to identify children and hazardous objects within the surrounding environment. Deep learning-based object detection models, such as YOLO, are intended to detect multiple household hazards including dangerous tools, stair edges, swimming pool boundaries, and cleaning chemical storage areas in real time.

In addition, the proposed system utilizes **Object Tracking** concepts to monitor the child's movement across sequential video frames. This helps the system observe the child's position and movement relative to surrounding hazards over time.

The system also incorporates **Spatial Analysis** and proximity measurement techniques to analyze the spatial relationship between the child and detected hazards. By estimating the distance between the child and dangerous areas or objects, the system can recognize potentially unsafe situations.

Furthermore, the proposed solution applies **Rule-Based Prediction logic** to support proactive alert generation. The system may analyze factors such as the child's proximity to hazards and movement toward dangerous zones in order to determine whether an early warning should be generated.

As illustrated in **Figure 5.1** the proposed **WALI** system follows a sequential AI workflow that integrates object detection, object tracking, spatial risk analysis, and real-time alert generation.

These integrated AI techniques collectively support the system's ability to perform intelligent monitoring, recognize unsafe situations, and assist caregivers through real-time safety alerts.



Figure 5.1: *AI Techniques and Processing Workflow in the WALI System*

5.2 Architectural Strategies

5.2.1 Core Algorithm

The proposed *WALI* system follows a sequential AI-driven pipeline designed to support real-time child safety monitoring within domestic environments. The system operates through multiple interconnected processing stages, beginning with live video acquisition and ending with proactive alert generation. To support efficient real-time operation, multiple YOLO model versions are comparatively evaluated, as presented in **Table 5.2**.

A. YOLO Model Evaluation and Selection: Different YOLO model versions are analyzed to assess their suitability for real-time hazard detection scenarios. The evaluation aims to identify a model configuration capable of balancing detection performance with real-time processing constraints. The implementation primarily considers Nano (n) variants due to their lightweight architecture and suitability for deployment on limited hardware resources.

Comparison Criteria	YOLOV5	YOLOV8	YOLOV10
Release Year	2020	2023	2024
Detection Speed	High	Very High	Fastest
Recognition Accuracy	Good	Excellent	Highly Advanced
Complexity	Simple	Balanced	High
Resource Usage	Low	Medium	Very Low

Table 5.2: *YOLO Versions Comparison*

B. System Logic and Implementation Phases

1. Live Stream Acquisition and Processing:

The algorithm starts by capturing a live video feed from the camera and decomposing it into sequential frames. Each frame is pre-processed for immediate analysis, ensuring the system responds within fractions of a second.

2. Detection and Element Identification:

The selected YOLO model is deployed to simultaneously identify the "Child" and "Domestic Hazards". The system generates Bounding Boxes around each detected element to determine its precise location within the scene.

3. Movement Tracking:

The system monitors the child's movement across sequential frames. This tracking mechanism helps in understanding the trajectory of the motion, distinguishing between normal movement and an active approach toward a hazardous area.

4. Proximity Analysis and Rule-Based Engine:

The algorithm estimates the distance between the child's bounding box and the hazard's bounding box. A Rule-Based Engine then evaluates this data:

Example: If the distance falls below a specific threshold and the movement vector indicates a direction toward the hazard, a "High-Risk" state is triggered.

5. Risk Assessment and Decision Making:

The risk level is evaluated based on the outputs from the rule-based engine combined with the direction of movement. If the system confirms that the child is in an unsafe situation, it immediately proceeds to the warning stage.

6. Proactive Alert Generation:

Upon identifying a real threat, the system generates an early warning and sends an immediate notification to the parents. If no danger is detected, the system remains in a continuous analysis state to ensure persistent protection.

5.2.2 Layered Architecture

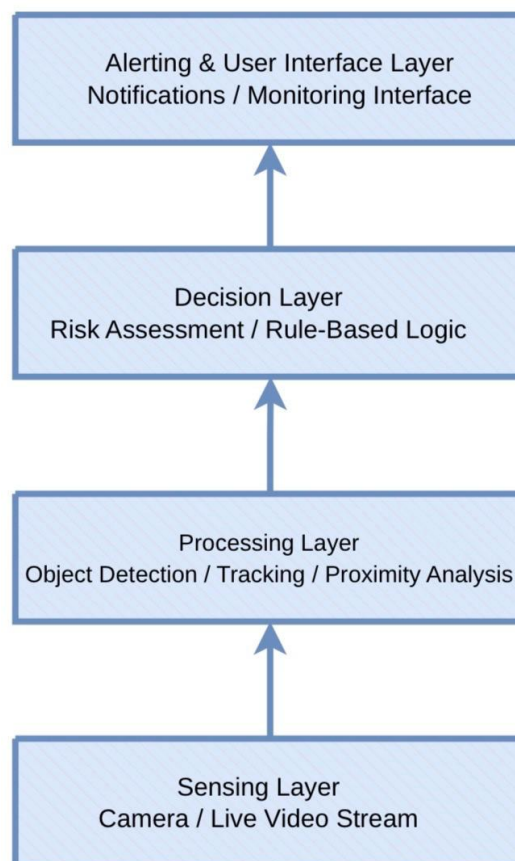


Figure 5.2: Layered Architecture of the Proposed *WALI* System

Figure 5.2 presents the layered architecture of the proposed *WALI* system, which is organized into four functional layers responsible for data acquisition, processing, decision making, and alert generation:

- 1) **Sensing Layer:** The layer responsible for initial data acquisition, capturing a live video stream of the domestic environment using camera sensors.
- 2) **Processing Layer:** Acts as the system's brain, where the YOLO model is executed to detect the child and hazards and determine their locations within the image.
- 3) **Decision Layer:** Responsible for intelligent analysis, utilizing a rule-based engine to evaluate the child's proximity to hazards and determining if the situation is safe or dangerous.
- 4) **Alerting and User Interface Layer:** The final output stage responsible for generating proactive alerts and sending immediate notifications to parents when a hazard is detected.

5.2.3 Future Enhancements

While the current *WALI* system provides a functional foundation for proactive child safety monitoring, several enhancements are planned to extend its capabilities in future development phases.

Expanded Hazard Detection Future versions will incorporate additional hazard categories beyond the current scope, such as electrical outlets, gas stoves, and open windows at height. This will require collecting supplementary annotated datasets and retraining the detection model to accommodate new classes.

IoT Integration A key planned enhancement involves integrating *WALI* within smart home ecosystems, allowing the system to interface with connected devices such as smart locks and appliance controllers. This would enable automated physical responses alongside caregiver alerts, shifting the system beyond notification into active intervention.

Multi-Camera Coordination Future development will address multi-room coverage through coordinated multi-camera tracking, ensuring consistent child monitoring across different viewpoints and eliminating coverage blind spots.

Adaptive Learning By systematically collecting feedback from deployed environments, the model can be periodically fine-tuned to reflect each household's specific conditions, reducing false alerts and improving long-term accuracy.

Edge AI and Offline Operation To enhance reliability, future iterations aim to migrate all inference and alert logic fully to edge devices, ensuring continuous operation without dependency on network connectivity while strengthening data privacy.

5.3 Project Management

Effective project management is a fundamental component in ensuring the successful transition of *WALI* from a conceptual framework to a fully operational safety system. This section outlines the structured methodology used to manage the project's lifecycle, ensuring that all technical and operational milestones were achieved within the specified timeframe. The management approach focuses on task synchronization, resource allocation, and the integration of hardware and software components through a well-defined development schedule.

Gantt Chart

Task ID	Task Name	General Description
Task 1	Requirement Analysis	Defining project objectives, scope, and functional requirements.
Task 2	Data Acquisition	Gathering and organizing the necessary datasets and information.
Task 3	Hardware Integration	Setting up the camera and physical components for system monitoring.
Task 4	System Implementation	Developing the core logic and functionalities of the monitoring system.
Task 5	Platform Integration	Connecting software components with hardware for seamless operation.
Task 6	User Interface Design	Designing the visual dashboard for real-time interaction and feedback.
Task 7	Final Documentation	Completing the technical report and preparing the final delivery.

Table 5.3: Description of Project Tasks

GANTT CHART

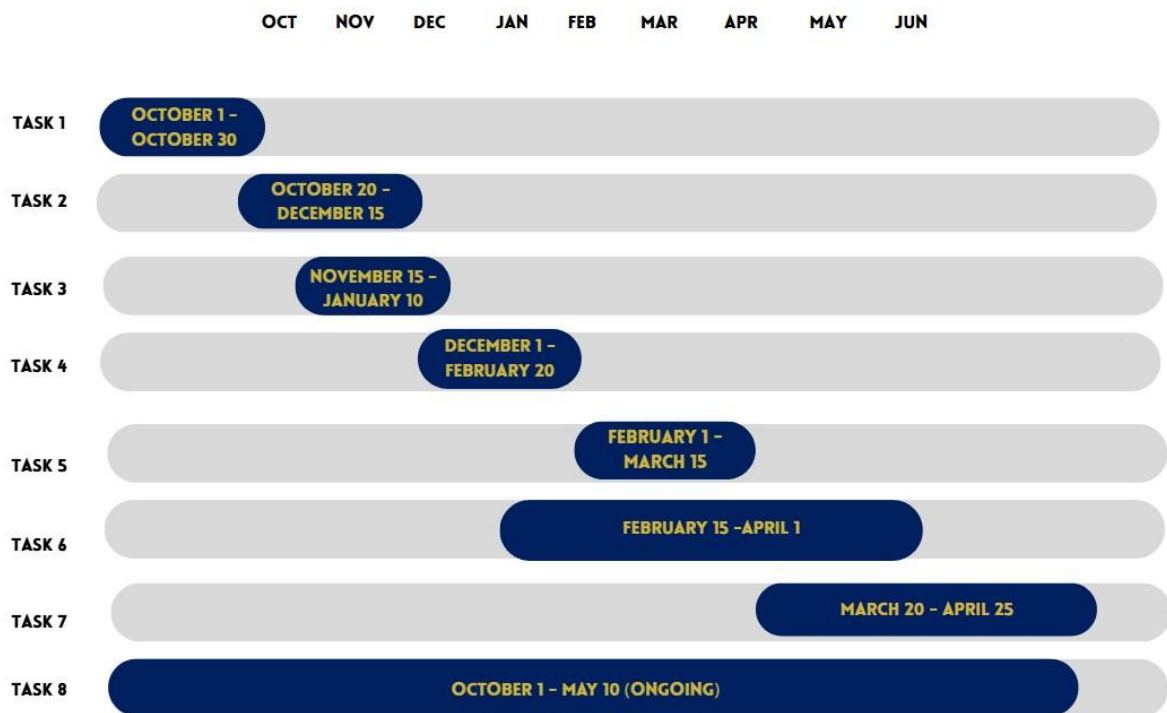


Figure 5.3: Project Gantt Chart

5.3.1 Risk Management

The development of the *WALI* system involves navigating various technical and operational uncertainties. Risk management is integrated into the project lifecycle to proactively identify, evaluate, and mitigate challenges that could compromise system integrity. This process focuses on environmental constraints—specifically occlusion and lighting variations—which are critical in computer vision applications. By implementing a structured mitigation framework, the project ensures high reliability and minimizes the rate of false alerts in sensitive monitoring environments.

Risk Management- Table

Risk Factor	Description & Impact	Mitigation Strategy	Risk Level
Environmental Constraints	Dynamic lighting conditions (glare or shadows) affect detection accuracy.	Deployment of robust image augmentation during training and adaptive thresholding.	Medium
Object Occlusion	Partial or full obstruction of monitored subjects by household objects.	Optimizing camera viewpoints and utilizing temporal tracking to maintain object ID.	High
System Reliability	Potential software latency or unexpected system failure during operation.	Implementing a modular software architecture and automated watchdog timers for recovery.	Low
Detection Fidelity	False positives/alerts caused by background noise or non-target movement.	Utilizing high-resolution datasets and refining the confidence score thresholds (mAP optimization).	Medium
Hardware Continuity	Connectivity issues or power instability affecting the monitoring feed.	Identification of redundant hardware components and rigorous pre-deployment testing.	Low
Human Factor	Over-reliance on the system or incorrect hardware placement by the user.	Providing a comprehensive user guide and clear system limitations notice.	Low

Table 5.4: *Comprehensive Risk Assessment and Mitigation*

5.3.2 Ethical Considerations

The development and deployment of *WALI* involve the continuous capture of visual data within private domestic environments, requiring developers to carefully address the associated ethical responsibilities.

Privacy Protection: All video streams processed by the system are intended for real-time analysis only and are not stored, shared, or transmitted to external servers without explicit user consent. The system is designed to operate locally wherever possible, ensuring that sensitive household data remains within the home environment.

Data Security: Access to the monitoring dashboard is restricted to authorized caregivers only, through secure authentication mechanisms. This ensures that live feeds and alert logs are not accessible to unauthorized individuals.

Child Data Sensitivity: Given that the primary subjects of monitoring are children, special care is taken to ensure that visual data is used solely for safety purposes and is never repurposed, shared, or used for any form of profiling or commercial exploitation.

Informed Consent: It is assumed that all individuals within the monitored environment are aware of the system's presence and have provided appropriate consent for its use.

Human Oversight: *WALI* is designed to support and assist caregivers rather than replace human judgment. All alerts generated by the system are intended to prompt human intervention, ensuring that final decisions regarding child safety remain with the caregiver.

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